

Experiment ‘p_2_6_convert’ Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_2_6_convert

Natural language statement: Write a method that takes a measurement in meters and then converts it to miles, feet, and inches.

Method signature: p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)

Ensures

- `x / 0.000621371 == miles`
- `x / 3.28084 == feet`
- `x / 39.3701 == inches`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a measurement in meters and then converts it to miles, feet,
↪ and inches.

The signature should be:

`p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)`

The method should respect the following contract:

ensures `x / 0.000621371 == miles`, ensures `x / 3.28084 == feet`, ensures `x / 39.3701 ==`
↪ `inches`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmdXnMCqOMjI937jJwgEctxeUfHZL

Model: gpt-5.5-2026-04-23

Created at: 1780484123

Finish reason: stop

Usage: 338 tokens in, and 190 tokens out

```
{
  miles := x / 0.000621371;
  feet := x / 3.28084;
  inches := x / 39.3701;
}
```

Final Program

```
method p_2_6_convert(x: real) returns (miles: real, feet: real, inches: real)
  ensures x / 0.000621371 == miles
  ensures x / 3.28084 == feet
  ensures x / 39.3701 == inches
{
  miles := x / 0.000621371;
  feet := x / 3.28084;
  inches := x / 39.3701;
}
```

Total Token Usage

Input tokens: 338

Output tokens: 190

Reasoning tokens: 133

Sum of ‘total tokens’: 528

Experiment Timings

Overall Experiment started at 1780484123148, ended at 1780484131028, lasting 7880ms (7.88 seconds)

Iteration #1 started at 1780484123148, ended at 1780484131028, lasting 7880ms (7.88 seconds)